

The Chef's Reference Library

Quick-access reference for the whole program. Print it, tab it, and keep it on the counter. Safety temperatures are drawn from the USDA Food Safety and Inspection Service and FoodSafety.gov.

1. The Standard Knife Cuts

Cut	Description	Typical use
Slice	Even flat pieces; thickness to taste	Tomatoes, onions, mushrooms
Rondelle	Round coin slices of cylindrical veg	Carrots, cucumbers, leeks
Julienne	Thin matchsticks ~ 1/8 in x 1/8 in x 2 in	Carrots, peppers, zucchini
Batonnet	Larger sticks ~ 1/4 in x 1/4 in x 2.5 in	Fries, crudite
Brunoise	Fine dice ~ 1/8 in cube (from julienne)	Aromatics, garnish
Small dice	~ 1/4 in cube (from batonnet)	Mirepoix, salsa
Medium dice	~ 1/2 in cube	Roasting vegetables, stews
Large dice	~ 3/4 in cube	Stocks, hearty stews
Chiffonade	Ribbons of leafy greens/herbs	Basil, mint, spinach
Mince	Very fine, irregular cut	Garlic, shallot, ginger

2. The Five Mother Sauces

Mother sauce	Base + thickener	Common derivatives
Bechamel	Milk + white roux	Mornay (+cheese), Soubise (+onion), Cream sauce
Veloute	White stock + blond roux	Supreme, Allemande, Normandy
Espagnole	Brown stock + brown roux + tomato	Demi-glace, Bordelaise, Chasseur
Tomato	Tomatoes + stock/aromatics	Creole, Marinara, Puttanesca-style
Hollandaise	Egg yolk + butter emulsion + acid	Bearnaise, Maltaise, Choron

Remember roux: equal parts fat and flour by weight. Cook it longer for a darker color and nuttier flavor, but darker roux thickens less.

3. Cooking Methods at a Glance

Method	Heat type	How it works	Best for
Saute	Dry	High heat, little fat, food moves	Quick-cooking vegetables, thin cuts
Sear / Pan-fry	Dry	Medium-high, develop crust & fond	Steaks, chops, fish fillets
Roast / Bake	Dry	Hot oven, dry convection	Whole birds, vegetables, breads
Grill / Broil	Dry	Radiant heat, char & smoke	Steaks, kebabs, vegetables
Deep-fry	Dry (in fat)	Submerge in hot oil 325-375 deg F	Fries, fritters, chicken
Steam	Moist	Cook in vapor, gentle	Fish, dumplings, vegetables
Poach	Moist	Gentle liquid 160-180 deg F	Eggs, fish, fruit
Simmer / Boil	Moist	185-212 deg F	Pasta, grains, stocks
Braise / Stew	Combination	Sear then slow-cook in liquid	Tough cuts, short ribs, beans
Sous vide	Moist	Precise low-temp water bath	Steaks, eggs, delicate proteins

4. Safe Minimum Internal Temperatures (USDA)

Food	Safe minimum	Notes
Poultry (all, whole & ground)	165 deg F / 74 deg C	Hold; juices run clear
Ground meats (beef, pork, lamb)	160 deg F / 71 deg C	No rest required
Beef, pork, lamb, veal (steaks/chops/roasts)	145 deg F / 63 deg C	Rest 3 minutes
Fish & shellfish	145 deg F / 63 deg C	Opaque, flakes easily
Egg dishes	160 deg F / 71 deg C	Cook until set
Leftovers & casseroles	165 deg F / 74 deg C	Reheat thoroughly
Ham, fresh or smoked (raw)	145 deg F / 63 deg C	Rest 3 minutes

Always verify with an instant-read thermometer in the thickest part, away from bone. Source: USDA FSIS & FoodSafety.gov.

5. Steak Doneness (quality, chef's guide)

Doneness	Pull temp	Center
Rare	120-125 deg F	Cool red center
Medium-rare	130-135 deg F	Warm red center (chef favorite)
Medium	135-145 deg F	Warm pink center
Medium-well	145-155 deg F	Slightly pink
Well-done	155 deg F +	Little to no pink

These are culinary doneness targets, not safety minimums. For ground beef and any compromised cut, follow the USDA table above. Pull a few degrees early and let carryover finish the cook.

6. Essential Culinary Ratios

Preparation	Ratio	Notes
Vinaigrette	3 : 1	Oil to acid (by volume)
Roux (by weight)	1 : 1	Fat to flour
Cooked white rice (absorption)	1 : 1.5-2	Rice to water
Pasta water salt	~ 1 Tbsp / liter	Salt to water (salty like the sea)
Bread (baker's basic)	5 : 3	Flour to water (~60% hydration)
Pie dough (3-2-1)	3 : 2 : 1	Flour : fat : water (by weight)
Stock	Cover by ~1 in	Bones plus aromatics, cold water
Brine	~ 6% salt	By weight of water (1 cup salt / gal approx)
Custard (basic)	2 : 1	Liquid to egg (varies by set)
Cooked-to-dry beans	~ 3 : 1	Cooked yield from dry weight

7. Measurement Conversions

Unit	Equals	Metric / note
1 tablespoon	3 teaspoons	15 mL
1 fluid ounce	2 tablespoons	30 mL
1 cup	8 fl oz / 16 Tbsp	237 mL
1 pint	2 cups	473 mL
1 quart	4 cups / 2 pints	946 mL
1 gallon	4 quarts	3.79 L
1 pound	16 ounces	454 g
1 ounce (weight)	28 grams	~2 Tbsp water
1 stick butter	8 Tbsp / 4 oz	113 g
Oven: 350 deg F	moderate	177 deg C / Gas 4
Oven: 425 deg F	hot	218 deg C / Gas 7

8. Glossary of Essential Terms

Term	Meaning
Mise en place	'Everything in its place'; all prep done before cooking.
Mirepoix	Aromatic base of onion, carrot, celery (2:1:1).
Sofrito	Aromatic base in Latin/Spanish cooking (onion, garlic, pepper, tomato).
Fond	The browned bits left in a pan; the base of pan sauces.
Deglaze	To dissolve fond with liquid (wine, stock) into a sauce.
Maillard reaction	Browning of proteins/sugars that builds savory flavor.
Caramelization	Browning of sugars alone; sweet, nutty flavor.
Roux	Cooked paste of fat and flour used to thicken sauces.
Emulsion	Stable mix of fat and water-based liquid (e.g., hollandaise).
Blanch & shock	Brief boil then ice bath to set color and texture.
Sweat	Cook gently in fat without browning to soften.
Sear	Brown a surface quickly over high heat.
Braise	Sear then slow-cook partly submerged in liquid.
Reduce	Simmer to evaporate liquid and concentrate flavor.
Temper	Slowly raise the temperature of eggs/chocolate to avoid shock.
Carryover cooking	Residual heat keeps cooking food after it leaves the heat.
Nappe	Sauce thick enough to coat the back of a spoon.
Brunoise	Fine 1/8-inch dice.
Chiffonade	Thin ribbons of leafy herbs or greens.
Brigade	The classic hierarchy of kitchen stations and roles.
Stage	An unpaid working trial/internship in a professional kitchen.
Deglaze	Lifting fond from a pan with liquid.
Sachet / Bouquet garni	Bundled herbs/spices used to flavor stocks and braises.
Umami	The savory 'fifth taste' from glutamates (tomato, cheese, soy).